- Is robust and can handle client dropouts up to a threshold value

Although federated learning was developed and visualised as an enterprise solution, there are quite a few open source frameworks that help you explore it even if you have access to a single desktop. Let's take a quick look at them.

## TensorFlow Federated

TensorFlow is one of the most popular frameworks used for building machine learning and deep learning models. Its popularity is nestled in the ability to run TensorFlow models reliably on any device — from a custom server to a handheld IoT device — and its ease of integration and extensive support for tooling. TensorFlow Federated is an open source framework built to support research and development of machine learning models using federated learning on decentralised data. It allows developers and researchers to simulate existing model architectures as well as test novel architectures for federated learning. It has two main interfaces.

*Federated Core API:* This is a programming environment for developing distributed computations. It serves as a foundation for the Federated Learning API and is used by systems researchers

*Federated Learning API:* This high-level interface helps machine learning developers and researchers incorporate federated learning without having to look into the low-level details.

Further documentation for this framework is available at *https://www.tensorflow.org/federated.*

## PySyft + PyGrid

PySyft is a Python library that aids in developing federated learning models for the purpose of research. It uses differential privacy and encrypted communications. It works in tandem with existing deep learning frameworks such as TensorFlow and PyTorch. However, PySyft alone isn't sufficient to work on problems that involve communications over a network. This is where PyGrid comes in, as it helps to implement federated learning on a wide variety of devices and to deploy PySyft at scale.

Further documentation is available at *https://github.com/OpenMined/PySyft.*

## OpenFL

Developed by Intel Labs, OpenFL is a federated learning framework built to be flexible, extensible, and learnable. It splits nodes into collaborators and aggregators. Collaborators train the model on their local data and send updates. Aggregators use updates from collaborators to update the global model. Communication is done between nodes using mTLS, which warrants that each node holds a valid certificate. OpenFL allows customisations in the form of lossy or lossless compression, logging, data split and a variety of aggregation logic.

For further documentation on this framework, you can refer to *https://github.com/intel/openfl.*

## APPFL: Argonne Privacy-Preserving Federated Learning

APPFL is an open source framework that helps develop novel models aimed towards privacy-preserving federated learning. It provides the framework for users to develop neural networks on decentralised data with differential privacy. Apart from standard implementations and data sets, it allows users to build customised models using the PyTorch module. Models developed with APPFL can be run from the terminal, using notebooks or on the Google Cloud platform. A developer can also simulate runs on a single or clustered environment with serial or parallel execution.

For the complete documentation, you can go to *https://github.com/appfl/appfl.*

## Flower

Developed by Adap, Flower is a unified approach to federated learning, analytics and evaluation. It was developed to be scalable to work with a wide range of machine learning frameworks and devices. Flower is scalable to the tune of handling tens of millions of clients. It works efficiently with a multitude of deep learning frameworks such as Keras, TensorFlow, PyTorch, Numpy, and so on. It was developed to run with the same efficiency on all kinds of devices such as servers, laptops, mobiles and other IoT and edge devices. It helps researchers deploy their novel algorithms with ease.

Further documentation on this framework is available at *https://flower.dev/.*

Since its inception, researchers and machine learning developers have been experimenting with federated learning on a small scale with very minimal upfront infrastructure. The frameworks introduced in this article are some of the most well-documented and beginner-friendly solutions for understanding and implementing federated learning. END

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